

Measurements of geomagnetically induced current in a power grid in Hokkaido, Japan

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[1] There have been numerous reports showing that space weather affects power grids through a geomagnetically induced current (GIC). Generally, power grids consist of power lines connected to transformers, of which neutral points are directly grounded. The GIC flows into those transformers through the neutral points if geomagnetic variations cause a ground level potential. These currents can damage power grids, especially transformers. It has been tacitly assumed, however, that the effect of the GIC is minor in Japan because of the country's location at geomagnetically lower latitudes. To examine the GIC effect in Japan, we conducted approximately 2 years of GIC measurements in Hokkaido, Japan. It is found that GICs associated with substorms can be detected in Japan even at the solar minimum although intense GICs do occur mostly during geomagnetic storms. Temporal variations of GICs show high correlation with geomagnetic field variations, rather than time derivatives of the geomagnetic field.

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1. Introduction

[2] Space weather affects man-made technological systems, such as satellites, power grids, and global positioning system (GPS) [e.g., *Lanzerotti*, 2001]. It is crucial for power companies to estimate the risk from geomagnetically induced currents (GICs) [e.g., *Kappenman et al.*, 1981; *Kappenman*, 2000; *Boteler et al.*, 1989]. On 13 March 1989, for example, the power blackout of the Hydro-Quebec system in Canada was attributed to an intense geomagnetic storm [*Kappenman*, 1989, 2001; *Boteler*, 2001]. On 30 October 2003, a power blackout of the high-voltage transmission system occurred in southern Sweden during a space storm [*Pulkkinen et al.*, 2005].

[3] It is widely known that high-frequency changes in the electrorjet currents at high latitudes associated with intense auroral activities produce intense GICs [*Kappenman*, 2005]. Impulsive geomagnetic disturbances caused by

interplanetary shocks such as storm sudden commencements (SSCs) and sudden impulses (SIs) are also recognized as a potential driver for intense GICs. *Kappenman* [2003] noted that a large SSC on 24 March 1991 produced the largest GIC measured in the United States. It was also found that intense GICs were produced by geomagnetic disturbances driven by intensification of the ring current at low latitudes [*Kappenman*, 2004], motivating the present study to measure GICs at geomagnetically lower latitudes to estimate their effect on power grids. This paper reports the GIC measurements in Hokkaido, Japan for approximately 2 years from December 2005.

2. Measurements of GIC

[4] We measured electrical currents in a transformer of the 187 kV power line system at the Memanbetsu substation (indicated by the square in Figure 1) of Hokkaido Electric Power Co. Inc. The direction of the line is primarily southwestward and the length of the line is approximately 100 km.

[5] A neutral point for three-phase alternating currents of the transformer is directly grounded for protection of the power system. Therefore, the GIC flows to the ground through the neutral point. The current intensity at the neutral point is measured using a clamp ammeter. The sampling rate was one millisecond, and 1-sec average data were transferred to the National Institute of Information

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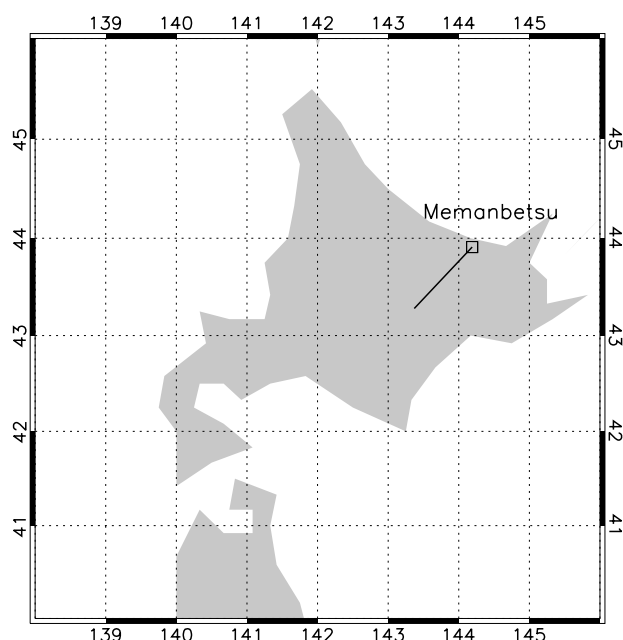


Figure 1. Configuration of measured 187-kV power line in the geographical coordinate system. The square shows Memanbetsu substation, where GICs were measured.

and Communications Technology (NICT) from the Memanbetsu substation for near real-time monitoring.

3. Data Analysis

[6] For our study we used 1-sec GIC data and 1-sec geomagnetic data, along with lists of the geomagnetic storms and bays obtained from the Memanbetsu Magnetic Observatory (MMB; geographic latitude: 43.9 N, geographic longitude: 140.2 E, geomagnetic latitude: 35.4 N, and geomagnetic longitude: 211.3 E), of the Japan Meteorological Agency (JMA) for the following analysis. We chose the

storms and the positive bays with distinct (A) and fair (B) levels in the lists.

[7] We generated a data set where we defined a GIC event as an hour interval during which the GIC intensity exceeds 1 A. We then performed a statistical analysis for the period between December 2005 and December 2007.

3.1. GIC Events and Geomagnetic Activities

[8] Table 1 is a list of geomagnetic storms with SSCs and gradual commencements (SGs) reported at MMB between December 2005 and December 2007. It also shows maximum values of GICs associated with these storms. Only 14 storms were observed during this period because of low solar activity near the solar minimum of Cycle 23. Intense GICs were observed during the geomagnetic storms. The most intense GIC was observed in association with the geomagnetic storm of 14 December 2006. The second largest was observed with the storm of 9 November 2006, the third one was with the storm of 30 November 2006. An SI on 10 July 2006 produced the forth largest GIC.

[9] Our measurements indicate that GIC events occur in conjunction with various types of geomagnetic activities. According to the analysis based on the lists of the storms and bays from MMB, approximately 33% of the events occurred in association with midlatitude positive bays (i.e., substorms). This percentage increases by approximately 46% if we include an uncertain (C) level in the MMB lists. Approximately 40% of the GIC events occurred during storms and in particular the intense GIC events were often occurred during the storms.

3.1.1. GIC Event Associated With the Geomagnetic Storm of 14–15 December 2006

[10] As shown in Table 1, intense GIC events are frequently observed in association with the geomagnetic storms. A geomagnetic storm occurred at 1414 UT on 14 December 2006. The cause of this geomagnetic storm was the full halo coronal mass ejection (CME) associated with the X3.4/4B flare (S06W24) at 0214 UT on 13 December.

Table 1. List of Geomagnetic Storms Reported From Memanbetsu Magnetic Observatory, Japan Meteorological Agency Between December 2005 and December 2007^a

Start Date	Start Time (UT)	End Date	End Time (UT)	ΔH (nT)	Type	Maximum GIC (A)
4 Apr. 2006	07.6*	6 Apr. 2006	1600	132	SG	1.29
8 Apr. 2006	22.3*	10 Apr. 2006	1800	106	SG	0.95
14 Apr. 2006	01.6*	16 Apr. 2006	1700	157	SG	1.58
27 Jul. 2006	1353	28 Jul. 2006	1700	130	SSC	1.06
7 Aug. 2006	0035	7 Aug. 2006	2100	97	SSC	1.29
19 Aug. 2006	10.5*	20 Aug. 2006	1900	103	SG	1.52
9 Nov. 2006	13.0*	11 Nov. 2006	2100	143	SG	2.23
30 Nov. 2006	02.4*	30 Nov. 2006	2200	108	SG	1.75
5 Dec. 2006	22.2*	7 Dec. 2006	1800	101	SG	0.78
14 Dec. 2006	1414	15 Dec. 2006	2400	272	SSC	3.85
10 Jul. 2007	18.1*	11 Jul. 2007	1000	100	SG	0.98
20 Jul. 2007	0616	20 Jul. 2007	1900	86	SSC	0.66
20 Nov. 2007	0908	21 Nov. 2007	1700	105	SSC	1.13
17 Dec. 2007	0252	17 Dec. 2007	2400	103	SSC	0.77

^aThis table is based on the list available at MMB. Start time of the SSC storms is determined using SSC time, while that of the SG storms is determined using approximate start time of the decrease of the geomagnetic field. The asterisks are added for start time of SG storms to express this.

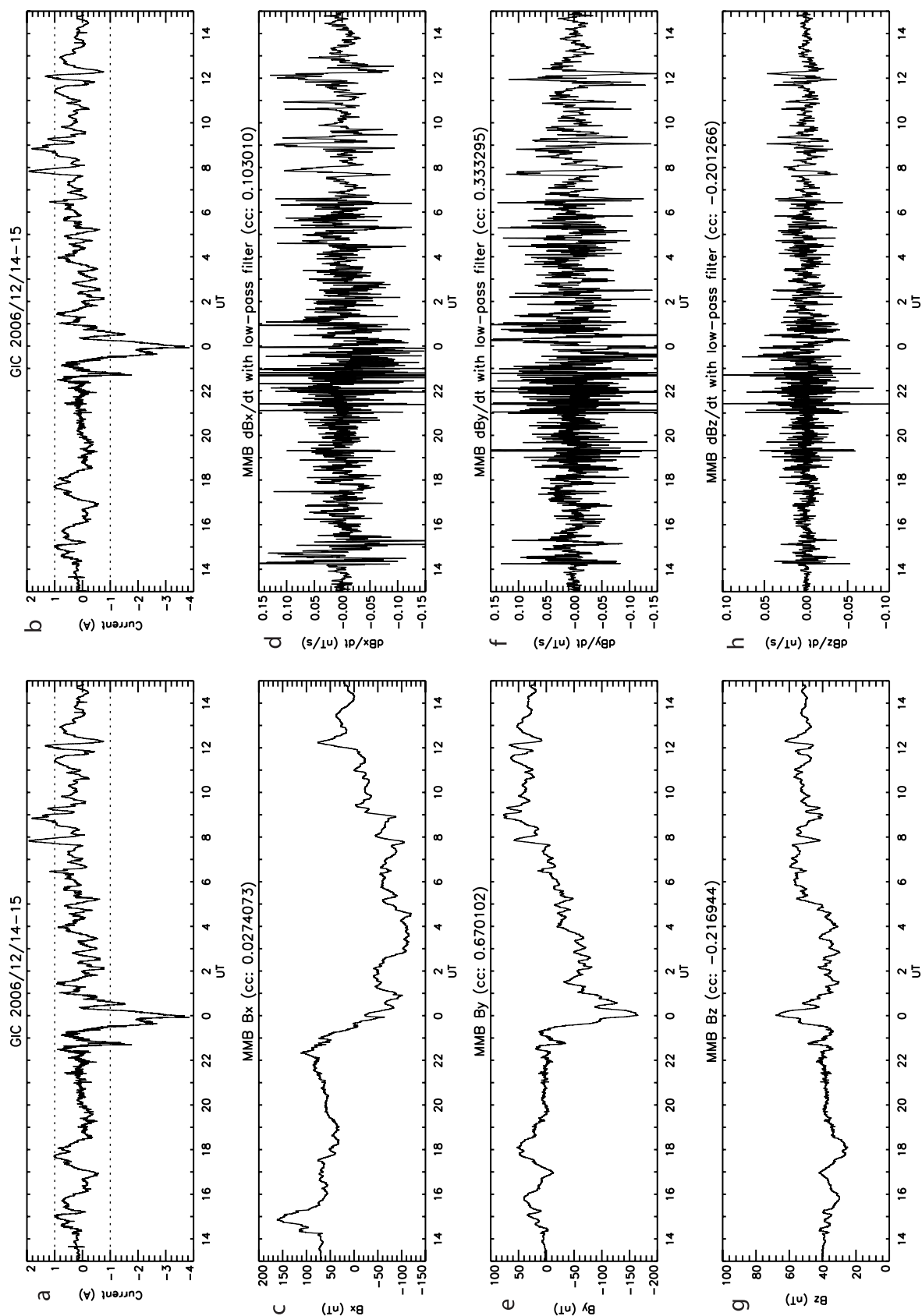


Figure 2. (a and b) GIC event associated with the geomagnetic storm of 14–15 December 2006 and (c) Bx, (e) By, (g) Bz, (d) dBx/dt, (f) dBy/dt, and (h) dBz/dt at MMB.

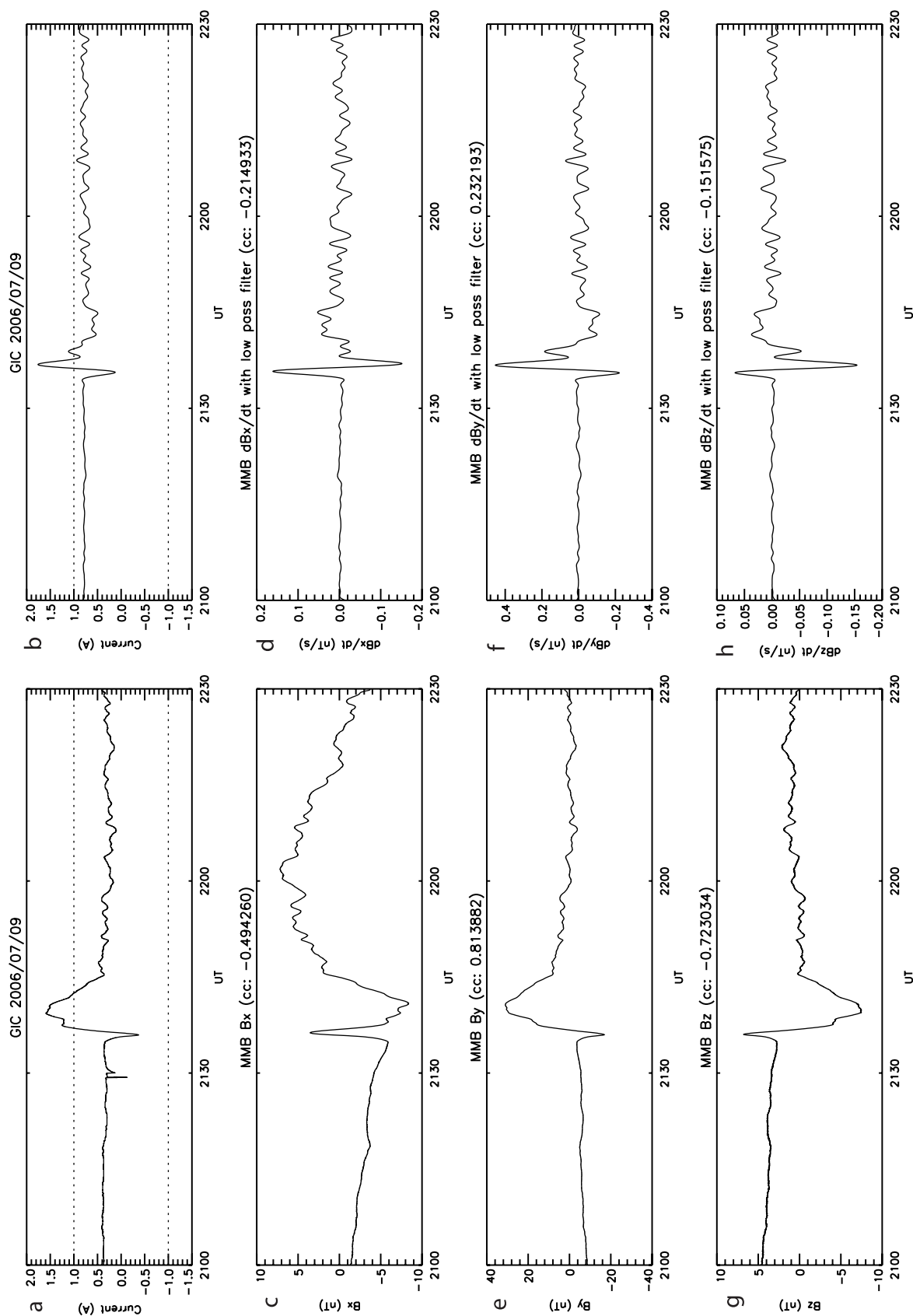


Figure 3. Same as Figure 2, except for (a and b) GIC event associated with an SI at 2134UT on 9 July 2006 and the (c) Bx, (e) By, (g) Bz, (d) dBx/dt, (f) dBz/dt, and (h) dBz/dt components of geomagnetic variations at MMB.

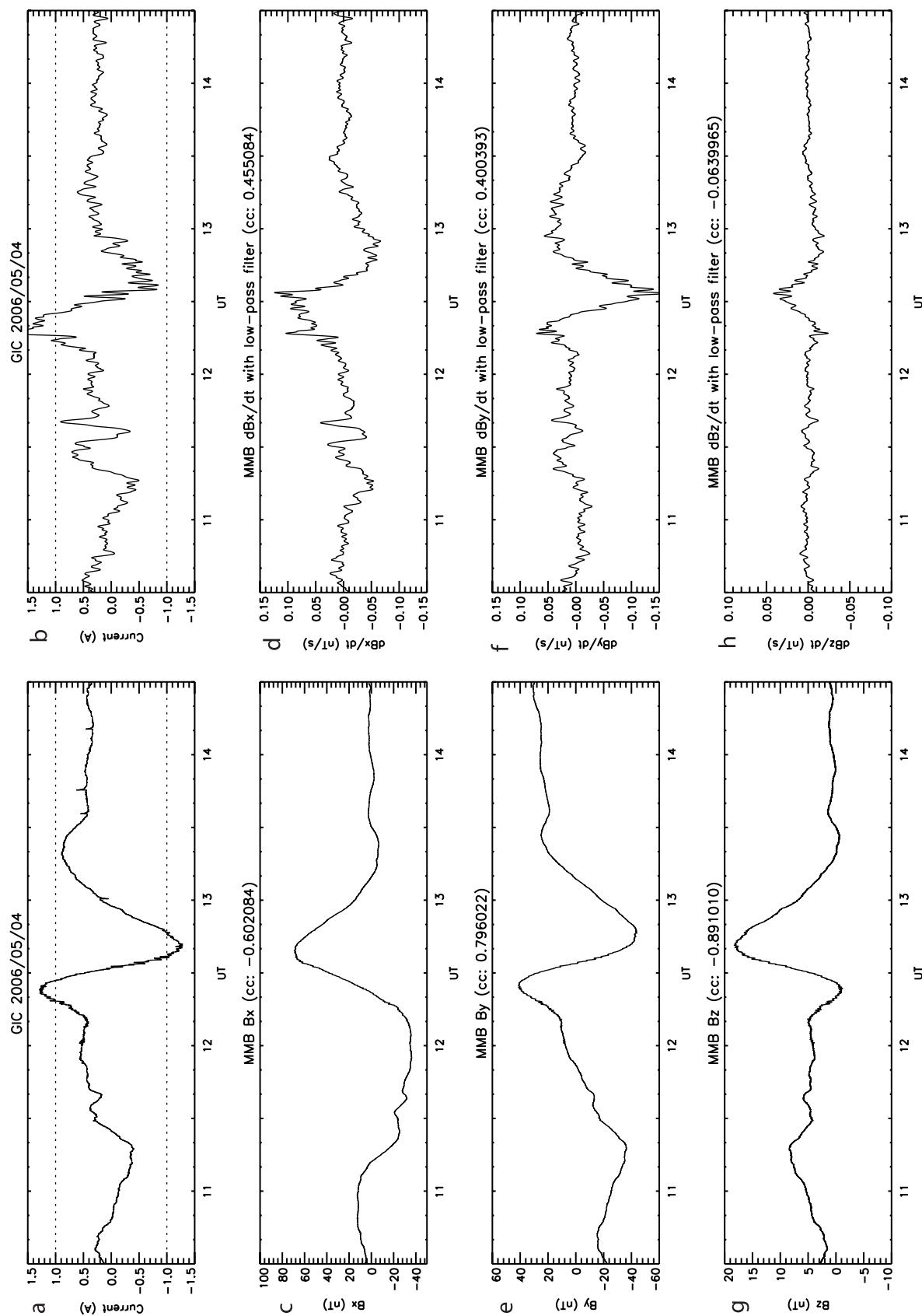


Figure 4. Same as Figure 2, except for (a and b) GIC event associated with a middle-latitude positive bay on 4 May 2006 and (c) Bx, (e) By, (g) Bz, (d) dBx/dt, (f) dBBy/dt, and (h) dBz/dt components of geomagnetic variations at MMB.

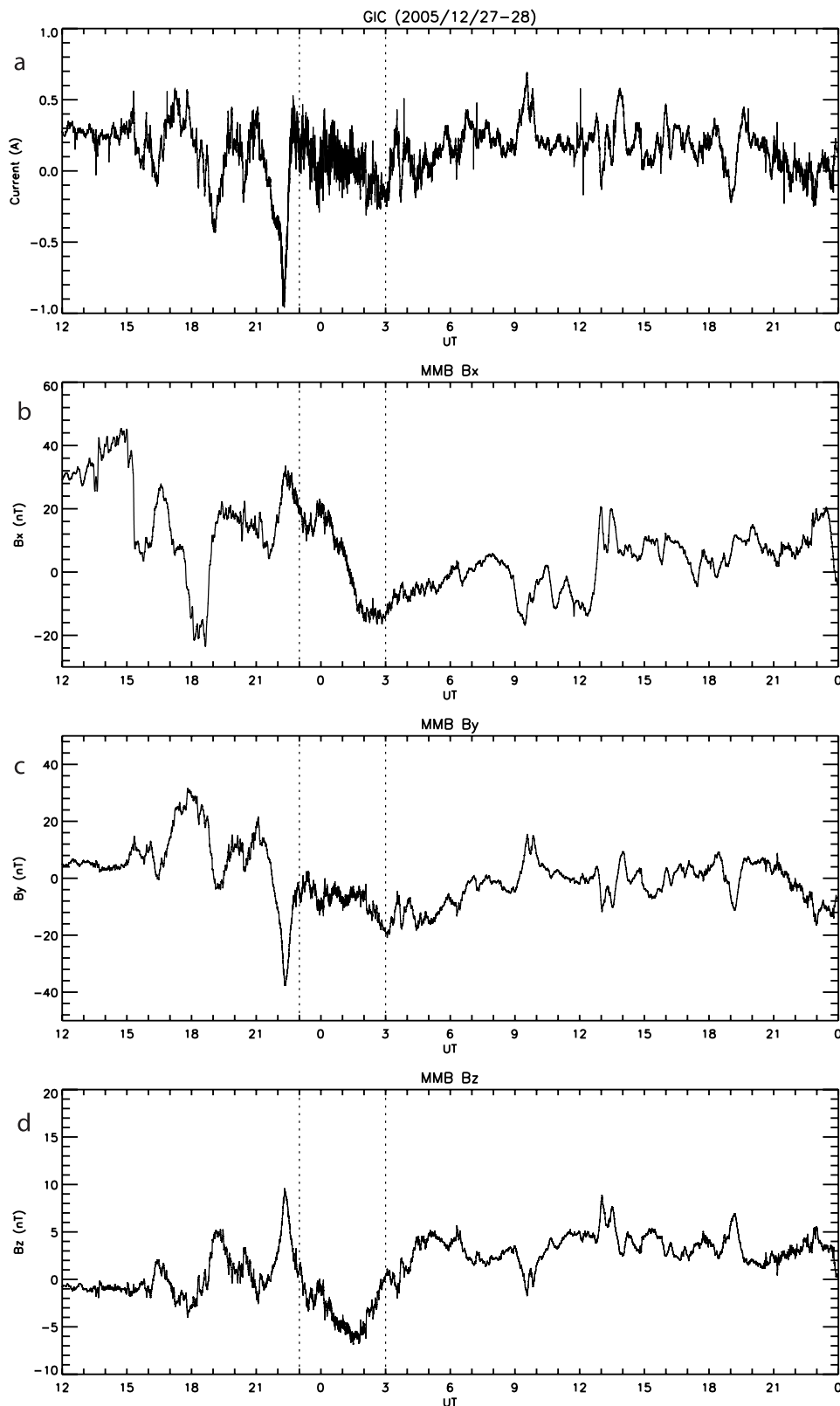


Figure 5. (a) GIC event associated with geomagnetic pulsations on 27–28 December 2005 and the (b) Bx, (c) By, and (d) Bz components of geomagnetic variations at MMB.

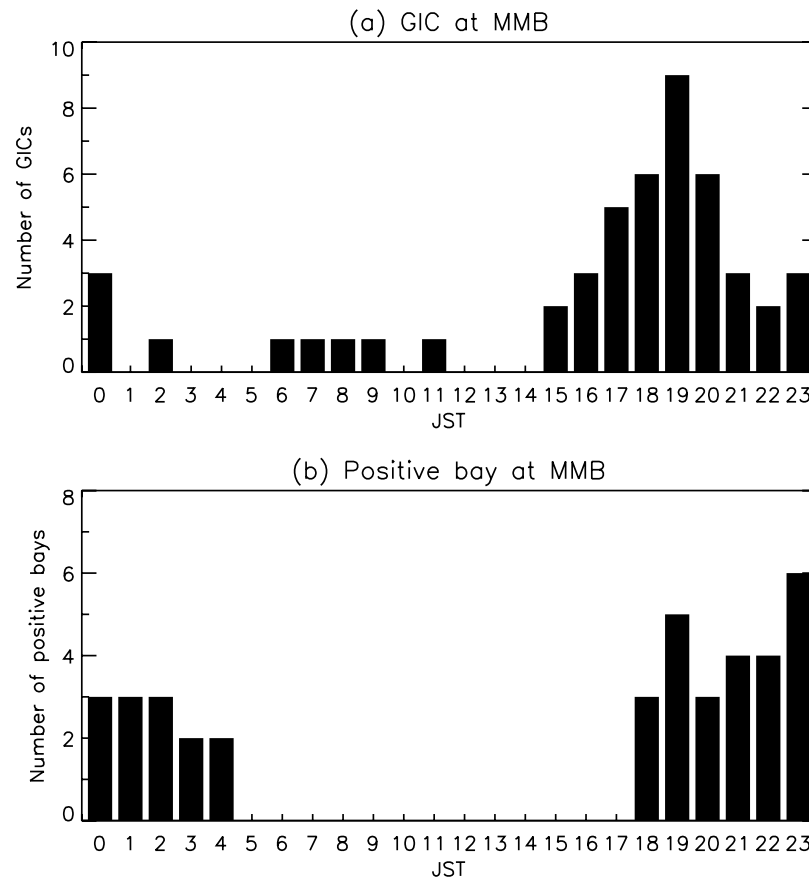


Figure 6. (a) Local time dependence of GIC events and (b) positive bays reported from MMB.

This flare was well covered by the observation of the Hinode satellite which revealed fine structures of this two-ribbon “white light” flare by a series of high-resolution images of the Hinode Solar Optical Telescope [Isobe *et al.*, 2007]. The disturbance took only 36 h to travel from the Sun to the Earth.

[11] Figure 2 shows the GIC variations during this storm, which is the most intense GIC case, together with the geomagnetic variations in the north–south (B_x), east–west (B_y), and vertical (B_z) components at MMB and the time derivatives of B_x (dB_x/dt), B_y (dB_y/dt), and B_z (dB_z/dt). The horizontal dotted lines in the top panels show ± 1 A levels of GICs. The time derivatives are smoothed using a 0.01 Hz low-pass filter. The correlation coefficients with GICs are 0.027 for B_x , 0.670 for B_y , -0.217 for B_z , 0.103 for dB_x/dt , 0.333 for dB_y/dt , and -0.201 for dB_z/dt . The GIC has the highest correlation with the B_y component.

[12] The B_x component changed greatly at the SSC at 1414 UT on 14 December, but the GIC corresponding to this SSC was not so large. This is understandable by considering that the change in the B_y component for this SSC was small, compared with that in the B_x component. The GIC with a peak value of 1.02 A was measured in association with substorm activities around 1500 UT

following this SSC. The most intense GICs were attributed to the rapid decrease in the geomagnetic field during the main phase at around 0000 UT, before this storm gradually recovered. The high cadence of substorm activities caused strong GICs during the recovery phase.

3.1.2. GIC Event Associated With Sudden Change of Geomagnetic Field on 9 July 2006

[13] GICs can be caused by sudden changes in the geomagnetic field such as SSCs and SIs originated from interplanetary shocks [Kappenman, 2005]. Figure 3 is an example of a GIC associated with an SI event observed at 2134 UT on 9 July 2006. In this case the correlation coefficients with the GIC for the period of 2132–2145 UT are -0.494 for B_x , 0.814 for B_y , -0.723 for B_z , -0.215 for dB_x/dt , 0.232 for dB_y/dt , and -0.152 for dB_z/dt . The B_y and B_z components of the geomagnetic field showed high correlation with the GIC.

[14] The cause of this SI was the arrival of the interplanetary shock from the partial halo CME. This CME originated from the M2.5/2F long-duration flare (S11W32) in AR0898 at 0823 UT on 6 July, accompanied with Type II and IV radio bursts. An enhancement of the solar energetic proton flux of energies higher than 10 MeV was observed by the GOES satellites.

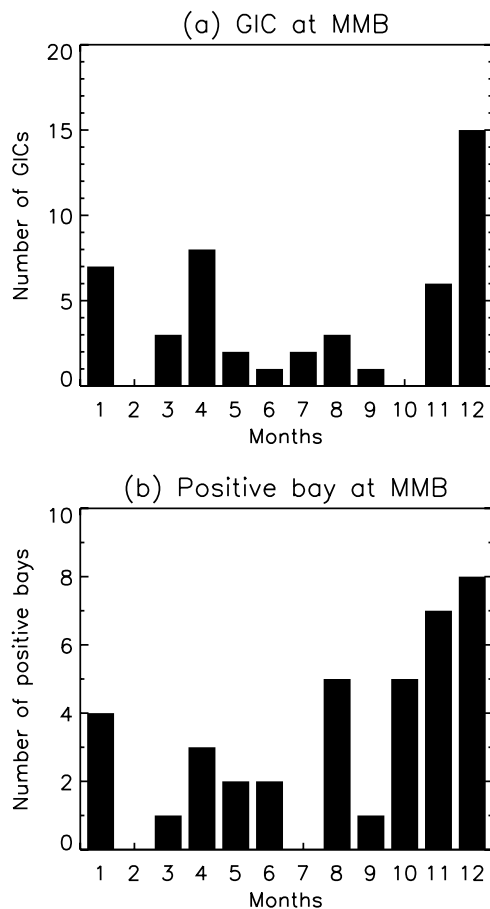


Figure 7. (a) Seasonal dependence of GIC events and (b) positive bays reported from MMB.

3.1.3. GIC Event Associated With Substorm Activity

[15] It is established that at high latitudes, GICs are produced by an increase in the electrojet currents [Boteler *et al.*, 1998; Kappenman, 2004]. At middle latitudes, on the other hand, GICs are often observed with “positive bays” as demonstrated by our measurements. It is usually the case that substorms produce positive bays at middle latitudes [Rostoker *et al.*, 1980].

[16] Figure 4 is a GIC produced by a substorm that took place around 1200 UT of 4 May 2006. The solar wind speed was slow: approximately 350 km. However, the interplanetary magnetic field was directed southward for several hours before this substorm. The correlation coefficients with the GIC are 0.602 for Bx, 0.796 for By, -0.891 for Bz, 0.455 for dBx/dt, 0.400 for dBy/dt, and 0.064 for dBz/dt. Again the GIC shows the high correlations with the By and Bz components.

3.1.4. GIC Event Associated With a Geomagnetic Pulsation

[17] Short-period fluctuations of geomagnetic fields are called ultralow frequency (ULF) geomagnetic pulsations.

Pulsations are classified into five categories in terms of their period ranges [Hughes, 1994]. Small amplitude fluctuations of GICs associated with the ULF geomagnetic pulsations were observed by our measurements. Pulkkinen and Kataoka [2006] and Kataoka and Pulkkinen [2008] reported the occurrence of GICs in pipe lines produced by Pc3–5 range geomagnetic pulsations. Figure 5 is an example of a GIC produced by geomagnetic pulsations on 27–28 December 2005. This event occurred in association with fluctuations in the high-speed solar wind from a coronal hole. According to the analysis between 2300 and 0300 UT (between the vertical dotted lines in Figure 5), there was an increase in power near the 10–100 s range in the power spectrum of the GIC, and By and Bz components.

[18] The effect of the ULF geomagnetic pulsations on power grids seems to be small because of their small amplitude. However, their effect is still important because continuous GICs by the ULF pulsations might electrically erode metallic pipe lines.

3.2. Local Time and Seasonal Dependencies

[19] Figure 6 shows the local time dependence of GIC events and of the occurrence of middle-latitude positive bays at MMB. It is noticeable that the GIC events are observed frequently in evening hours with a peak around 1900 JST. There is also a slight increase in occurrence frequency near local midnight. There were peaks of the GIC events around 1900 and 2300 JST, although the occurrence frequency of the positive bays increases more near midnight than the GIC events.

[20] Figure 7 shows the seasonal dependence of the GIC occurrence and of the occurrence of positive bays at MMB. The dependence indicates that the number of positive

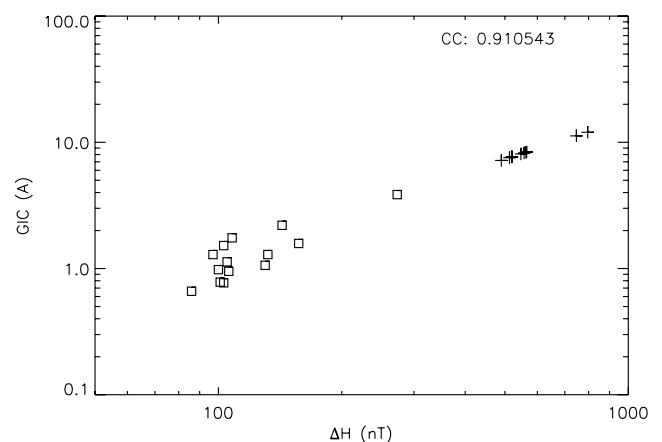


Figure 8. Scatterplot of maximum depressions of the horizontal component of the geomagnetic field (ΔH) of geomagnetic storms and maximum values of GICs. Squares show measured GICs in Table 1, and crosses show estimated values of GICs for 10 largest geomagnetic storms in Table 2.

Table 2. List of 10 Largest Geomagnetic Storms Reported From Memanbetsu Magnetic Observatory Since 1957

	Date	Time (UT)	Duration (hours)	ΔH (nT)	SSC (nT)	Estimated Maximum GIC (A)
1	13 Jul. 1982	1617	—	796	148	12.0 ± 3.0
2	13 Mar. 1989	0127	68.6	747	94	11.2 ± 2.8
3	8 Jul. 1958	0748	51.2	565	165	8.4 ± 2.0
4	15 Jul. 1959	0802	42.0	563	92	8.3 ± 2.0
5	11 Feb. 1958	0126	52.6	557	50	8.2 ± 2.0
6	25 May 1967	1235	103.4	547	202	8.1 ± 1.9
7	15 Jul. 2000	1436	27.4	520	180	7.7 ± 1.8
8	5 Aug. 1972	1400	52.0	520	63	7.7 ± 1.8
9	29 Oct. 2003	0611	—	513	112	7.5 ± 1.8
10	12 Nov. 1960	1348	56.2	490	48	7.2 ± 1.7

bays also increased in April, November, and December. An increase of GIC events on December might be caused by the activity of December 2006 shown in section 3.1.1.

4. Discussion

[21] Our measurements have demonstrated that temporal variations of GICs show a high correlation with those of geomagnetic field (in the By and Bz components) rather than time derivatives of the geomagnetic field. *Lanzerotti et al.* [1995] pointed out a similar tendency for their GIC measurements in marine cables. *Trichenko and Boteler* [2006] also noted this and suggested the effect of underground conductivity structures. *Owada* [1972] reported that the subterranean electrical structure near Memanbetsu is composed of three layers. The electrical conductivities are 0.01–0.03 S/m in the first layer (8–20 km), 0.0002–0.001 S/m in the second layer (20–90 km), and 0.002–0.003 S/m in the third layer (90–170 km). *Uyeshima et al.* [2001] made the measurements of underground conductivity structures in eastern Hokkaido using a network-magnetotelluric method. It was then suggested that the coast effect [*Parkinson, 1959; Parkinson and Jones, 1979*] may practically explain their results.

[22] As other possibilities, *Kikuchi et al.* [1978] proposed that the zeroth-order transverse magnetic waveguide mode can account for the transmission of polar electric fields toward the equator. According to this waveguide mode, the GIC variations at middle latitudes can be explained reasonably by a return current in the ionosphere. The occurrence of GICs increases around local midnight at high latitudes associated with auroral activities [*Viljanen et al., 2001*]. Figure 6 shows that the occurrence of the positive bays increases near midnight, but our result shows an increase of GIC events in local evening, instead of midnight. The ionospheric conductivity becomes lower on the night side. A balance between the two causes determines the occurrence of GIC events. This was noted in the return current hypothesis of *Kikuchi et al.* [1978; see also *Motoba et al., 2002*].

[23] Regarding the seasonal dependence, it is known that geomagnetic storms tend to maximize in spring and fall [*Russell and McPherron, 1973; Crooker et al., 1992; Cliver et al., 2000*]. However, the seasonal dependence is not clear in our analysis. The reason for this is that number of

storms was small in our data set because of the only 2 years of measurements near the solar minimum.

[24] Our study has confirmed that GICs occur most often during geomagnetic storms. Figure 8 shows a scatter-plot of maximum values of GICs and maximum depressions of the horizontal component of the geomagnetic field (ΔH) of at the main phase of geomagnetic storms. There is a good linear correlation between ΔH and the maximum GIC intensity. The correlation coefficient is 0.91 when we used all data in Table 1 (shown by squares in Figure 8). The correlation coefficient is 0.62 if we excluded the storm of 14 December 2006. We have derived an empirical equation of the maximum intensity of GIC and ΔH at MMB for our data set, including the storm of 14 December 2006

$$\text{maximum GIC(A)} = 0.0158\Delta H(\text{nT}) - 0.558. \quad (1)$$

Table 2 shows the ten largest geomagnetic storms since 1957 reported from MMB and the estimated maximum values of GICs by those storms using equation (1). The estimated values are also plotted with crosses in Figure 8. The largest geomagnetic storm occurred on 13 July 1982. The maximum value of horizontal variation of this storm was 796 nT. According to our estimate, the maximum GIC value of 12.0 ± 3.0 A must be associated with the storm on 13 July 1982. Using Figure 13 of *Kappenman* [2004], we estimate the GIC value of approximately 45 A in central Japan power grid for the same storm. We think that this difference is produced by the effects of underground conductivities and configurations of power grids.

5. Summary

[25] We have reported the initial result on GICs from the approximately 2 years of measurements in Hokkaido, Japan. The GICs were observed in conjunction with various geomagnetic activities, e.g., geomagnetic storms, sub-storm activities, sudden changes of geomagnetic fields associated with interplanetary shocks, ULF geomagnetic pulsations, while the intense GICs were associated with geomagnetic storms. Our measurements indicated that GICs have better correlation with geomagnetic field variations rather than with time derivatives of the geomag-

netic field. We pointed out some possible mechanisms to explain this correlation. Further examinations are necessary to evaluate them. Although it is believed that power grid problems from GICs hardly occur in Japan because of the country's location at geomagnetically lower latitudes, our study will aid in preparing for the extreme events shown in Table 2.

[26] **Acknowledgments.** We would like to thank Hokkaido Electric Power Co. Inc. for allowing us to measure GICs at the Memanbetsu substation. We would also like to thank the Kakioka Magnetic Observatory of the JMA for providing us with 1-sec geomagnetic data and the lists of the geomagnetic storms and bays prepared by the MMB.

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